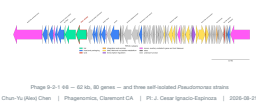
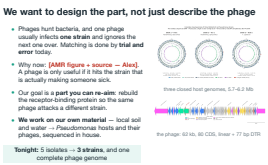
2026-08-21



Good evening. We are trying to build phage parts on purpose — to take the protein that decides which bacterium a phage can infect, and redesign it. Tonight: the two things you need before you can.

(≈18 s)

We want to design the part, not just describe the phage



Phages hunt bacteria, and they are picky: one phage infects one strain and ignores the next one over. Today that match is trial and error.

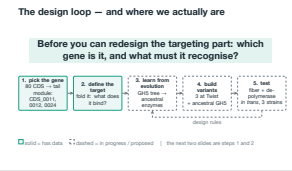
That matters right now — [AMR number, Alex fills in]. A phage is only useful if it hits the strain making someone sick.

So our goal is not to describe a phage; it is to build a part you can re-aim — rebuild the receptor-binding protein so the same phage attacks a different strain.

We work on our own material: local soil and water, Pseudomonas hosts and their phages, sequenced in house — five isolates, three strains, one complete phage genome.

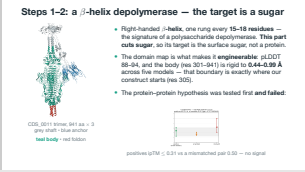
(≈48 s)

The design loop — and where we actually are



Here is the loop. Step one, pick the gene: of 80 genes the structural module narrows it to a handful — the tail spike, the GH5 beside it, and a depolymerase the annotator missed.
Step two, define the target — you cannot redesign a binder without knowing what it binds.
Step three is where evolution comes in. The tail spike decides who is recognised; the depolymerase decides what can be cut open. So we build a tree over all GH5 sequences and reconstruct ancestral enzymes — proteins from different points in evolution, not random mutants. Stated caveat: this assumes affinity tracks infection.
Step four, build them: three constructs are at Twist, ancestral enzymes next. Step five, combine fiber and depolymerase in trans on our three strains, and feed that back as design rules.
Solid boxes have data; dashed ones do not. The next two slides are steps one and two.
(≈62 s)

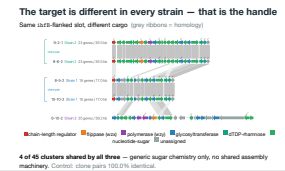
Steps 1–2: a β-helix depolymerase — the target is a sugar



Steps one and two on one slide. This is the tail spike, 941 residues, a trimer: grey shaft AlphaFold cannot place, blue anchor into the capsid, teal working body.
That body is a right-handed beta helix, one rung every 15 to 18 residues — the signature of a depolymerase. So the part cuts sugar, and its target is the surface sugar, not a protein.
It is also engineerable: the body from residue 301 is rigid, under one angstrom across five models, and that boundary is exactly where our construct starts.
And we tested the protein-protein hypothesis first — docking gave positives at 0.31 and a mismatch at 0.50. No discriminating power, so we retired that screen.

(≈57 s)

The target is different in every strain — that is the handle



If the part reads a sugar, does that sugar differ between strains? That difference is the handle a redesigned part would turn.

Five tracks: two Strain 2 clones, two Strain 1 clones, Strain 3; arrows are genes, grey ribbons are homology. Control first: inside each clone pair all 23 and all 15 genes are identical. ANI saturates, so this had to be measured.

Across strains the links are sparse and crossed — of 45 clusters only four are shared, all generic sugar chemistry, no shared flippase, polymerase or chain-length regulator.

Three strains, three targets. This phage should not plaque on the other two, and a re-aimed part is what would change that.

(≈57 s)

Where the loop stands



Where the loop stands. We know which gene to engineer and what it must recognise — the tail-spike depolymerase against a strain-specific O-antigen — and three strains give three targets.

The limitation is honest: steps three to five are not results — the GH5 tree is still being built, the constructs are not expressed, and the July fourth plaque data is not in our hands. And no serotyping tool exists for this taxon, so we cannot name an O-type.

Next three months: recover that readout, express the constructs, and reconstruct ancestral GH5 enzymes — then measure how proteins from different points in evolution behave with the fiber, in trans, on the three strains.

We know which part to rebuild and what it has to grip. The ancestral variants are the next lever.

(≈58 s — stop here, not on a thank-you slide)